MissingNo.Zero: pure self-play reinforcement learning reaches #1 on Pokémon Showdown in five random-battle generations

Technical report

Nick Greenquist

This is a personal project. Opinions and views are solely my own and not my employer's.

Abstract

In September and October 2026, MissingNo.Zero, trained *tabula rasa* by self-play reinforcement learning, reached #1 on Pokémon Showdown's public random-battle ladders in five generations: 1 (2026-09-29), 2 (2026-10-18), 4 (2026-10-08), 6 (2026-10-16) and 7 (2026-10-24). It trains a small policy-and-value network for each generation with PPO and plays with a short search used only at play time. The gen-1 agent trained on one laptop in about 40 hours, with no cloud compute.

1 Result

MissingNo.Zero reached #1 on five of Pokémon Showdown's random-battle ladders: generations 1, 2, 4, 6 and 7 (Figure 1). Tables 3 and 4 give each run's numbers.

2 Background

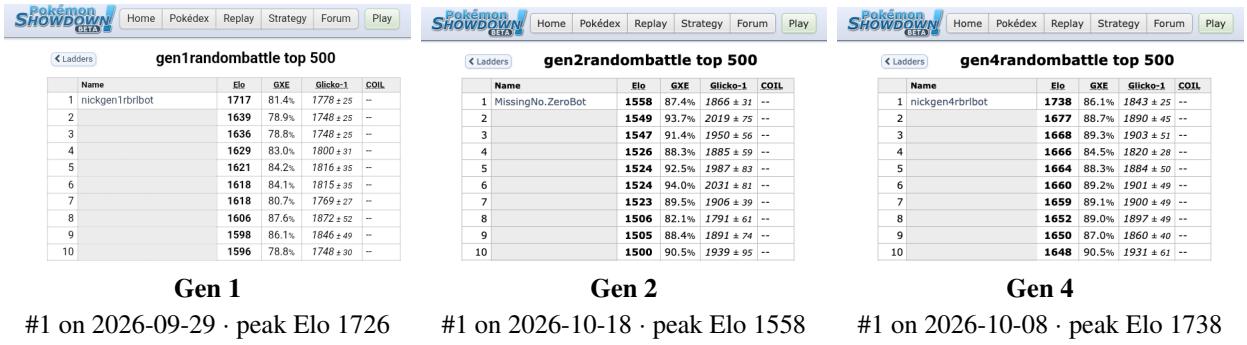
2.1 The game

Pokémon Showdown is a free online battle simulator with a public ladder for each format, ranked by Elo. Two players each have a team of six Pokémon, and each Pokémon has up to four moves. Every turn both players choose at the same time, either a move or a switch to another Pokémon, and a player wins when all of the opponent's Pokémon have fainted. In random battles each player is dealt a random team, so the game is the battle itself. Each generation of the games has its own rules, Pokémon and moves, so each is a different game with its own ladder.

2.2 Difficulty

Random battles combine several things that make games hard for learning agents:

- **Hidden information.** The opponent's Pokémon, and their moves, items and abilities, stay unknown until they are revealed, so the agent acts on its guesses about them.
- **Chance.** Damage rolls, accuracy, critical hits and side effects make each turn's outcome random, so learning what an action is worth takes many games.
- **Simultaneous moves.** Both players commit each turn without seeing the other's choice, so a predictable player can be exploited, as in rock-paper-scissors.
- **Size.** Random teams and many turns of branching make the game far too large to enumerate: Wang (2024) puts a conservative lower bound of about 10^{88} on the game tree of generation 4's random battles.



Gen 1
#1 on 2026-09-29 · peak Elo 1726

Gen 2
#1 on 2026-10-18 · peak Elo 1558

Gen 4
#1 on 2026-10-08 · peak Elo 1738

Gen 6
#1 on 2026-10-16 · peak Elo 1942

Gen 7
#1 on 2026-10-24 · peak Elo 1900

Gen 9
best #5 on 2026-10-25 · peak Elo 2453

Figure 1: Each board's top 10 with our account at #1 (gen 9's at #5, its best; it did not reach #1), at the highest Elo we have an image of (gen 1's shows 1717, below its 1726 peak). Other players' names are blanked.

3 Approach

We take AlphaZero's premise (Silver et al., 2018): one general algorithm that learns from self-play alone. Each Pokémon generation is a new game for the same algorithm; what we change per generation is the rules, the input encoding, and some training and search settings.

- **Self-play, no human data.** Each agent learns only from games against itself and its past versions: no human replays, no imitation of any player or bot, and no scripted opponents.
- **PPO.** A small policy-and-value network (about 3M parameters) is trained with proximal policy optimization (Schulman et al., 2017), with settings tuned per generation. Our starting recipe came from Wang (2024), whose gen-4 agent also paired PPO with play-time search. As in his work, every run learns from both players' sides of each self-play game.
- **Training simulator.** Gen 1 trained in pkmn/engine, an open-source re-implementation of the gen-1 rules that is fast and nearly exact. Gens 2, 4, 6, 7 and 9 trained on Showdown's own simulator, the program that runs the ladder, so they learned the ladder's exact rules. Every game on Showdown, in training, on the ladder and against Foul Play, reaches the agent through poke-env, an open-source Python client for Showdown.
- **Search only at play time.** Unlike AlphaZero, and as in Wang (2024), training uses no search. On the ladder, a short look-ahead search guided by the network's own policy and value picks each move, running on one laptop. For gens 2, 4, 6, 7 and 9 it plays moves out in poke-engine (pmariglia, 2025), an open-source re-implementation of the rules.
- **Measuring strength offline.** Ladder games are few, so we judged each model offline, in thousands of battles against Foul Play, an open-source search bot (pmariglia, 2025). Foul Play searches for a set time per move, so its strength depends on the machine; we fixed its search to the amount it completes in that time on our laptop, so it plays the same on any machine and evaluations could run in parallel on rented servers. Before each ladder run the new model and the previous one played Foul Play side by side.

3.1 What we leave out

We describe the approach only at a high level and release no code or weights for now. Showdown's bot policy is new, and we would rather not make it easier to run strong bots that do not follow it.

3.2 Training cost

Every network trained from random weights. Later generations rented one GPU at a time to go faster (Table 1).

Gen	Training steps	Hardware	Time to train	Rented compute
1	100M	laptop (Apple M4 Pro)	about 40 hours	none
2	600M	rented GPU	about 3 days	about $160
4	700M	rented GPU	about 6 days	about $230
6	600M	rented GPU	about 4 days	about $170
7	600M	rented GPU	about 4 days	about $220
9	800M	rented GPU	about 8 days	about $280

Table 1: Training cost.

3.3 Strength against Foul Play

Table 2 gives each model's win rate against Foul Play, without and with its play-time search, both played side by side.

Gen	Without search	With search	Battles per column
1	71.0% (69.3–72.6)	71.2% (69.6–72.8)	3,000
2	69.9% (64.0–75.2)	80.9% (75.6–85.2)	256
4	61.6% (59.4–63.7)	67.5% (65.4–69.5)	2,000
6	66.4% (60.4–71.9)	68.4% (62.4–73.7)	256
7	66.8% (60.8–72.3)	71.9% (66.1–77.0)	256
9	64.5% (58.4–70.1)	66.8% (60.8–72.3)	256

Table 2: Each model against Foul Play: win rate with its 95% Wilson interval. Foul Play's search is fixed per generation, so compare within a row only. For gens 2, 4 and 6, these games used a smaller search than the ladder did, so their with-search numbers likely understate ladder strength.

4 Ladder results

4.1 Each #1

Table 3 gives when each generation first reached #1, its rating and how many other players were active on the board; Table 4 gives how well it held #1 and the top 10 afterward.

Gen	#1 reached (UTC)	Peak Elo at #1	GXE at run end	Active opponent accounts, 24 h[a]
1	2026-09-29	1726	81.7%	85
2	2026-10-18	1558	87.0%	43
4	2026-10-08	1738	86%	36
6	2026-10-16	1942	91.7%	140
7[b]	2026-10-24	1900	89.5%	150

Table 3: Ranking. GXE is Showdown's estimate of the chance to beat a random ladder player. (a) Other accounts on the top-500 board that played at least one rated game in the 24 hours around the #1 (gen 1's starts about 5 hours after it). (b) Gen 7 reached #1 after Showdown made ladder games against bots opt-in (Section 4.4).

Gen	Longest streak of battles started at #1	Battles finished at #1 after the first #1[b]	Battles started in the top 10, after first reaching it
1[a]	23	23 / 75 (31%)	56 / 113 (50%)
2	2	2 / 30 (7%)	19 / 38 (50%)
4	15	17 / 23 (74%)	26 / 29 (90%)
6	23	33 / 43 (77%)	95 / 99 (96%)
7	2	1 / 19 (5%)	49 / 91 (54%)

Table 4: Holding #1. (a) Gen 1's run predates our hourly board snapshots, so its ranks were rebuilt from one board snapshot taken after the run; its streak and counts may each be one too high. (b) Of the battles after the agent first reached #1, how many it finished at #1.

4.2 How each #1 is documented

Each #1 is backed by Wayback Machine captures of the public board and the account's profile, taken while it was #1, and by git tags and file hashes made at the time, timestamped in Bitcoin with OpenTimestamps (Appendix A).

4.3 How the runs were conducted

Gens 1, 6 and 7 played their planned length (200, 100 and 100 battles). We stopped gen 4 after 87 battles, while it was #1 and Showdown was still setting its bot policy, and gen 2 after 46 battles (it first reached #1 at battle 16), because its board was thin and games were slow to find. Each account had played 150 to 800 rated games with earlier models before its run.

4.4 Bot policy

Every account had "bot" in its name, the agent never posed as a person, and no moderator contacted us during any run. Showdown opened a bot registry on 2026-10-13 and marked our accounts as bots, so gens 2, 6, 7 and 9 played as registered bots; gens 1 and 4 reached #1 before the registry existed.

Gens 1, 2, 4 and 6 played before 2026-10-20, when Showdown made ladder games against bots opt-in (off by default; smogon/pokemon-showdown de1cf598c). Until then the ladder paired a bot like any other

account, by rating, with whoever was searching for a game; being registered did not change that. Gens 7 and 9 played after the change, so they could meet only players who had opted in, and other bots. We do not know what share of the board's active players had opted in.

4.5 Generation 9

Gen 9's agent did not reach #1. Its best was #5, at Elo 2453 on 2026-10-25 (Figure 1). We stopped there rather than pay for a longer run. We did not ladder generations 3, 5 or 8.

5 Limitations

- **Formats.** Random-battle singles only; team building and doubles are untested.
- **Small boards.** A #1 is relative to who was playing: 36 to 150 other accounts were active in the 24 hours around each #1 (Table 3), and the strongest human players need not have been among them; gen 7 could be matched only with players who had opted in to bot games (Section 4.4).
- **Rank is not rating.** The board is ordered by Elo, which moves with recent streaks; how well each agent held #1 varied widely, from 1 of 19 battles in gen 7 to 33 of 43 in gen 6.
- **What mattered is unmeasured.** We changed the recipe between generations without ablations, so this paper does not show which parts of it were needed.

6 Related work

6.1 Self-play

AlphaZero (Silver et al., 2018) mastered chess, shogi and Go with one self-play algorithm, searching during training as well as play.

6.2 Jaxcalibur

Jaxcalibur reached #1 on Showdown's generation-9 random-battle ladder in August 2026 with self-play PPO and play-time search (self-reported; Jaxcalibur, 2026); its author released no code or weights.

Our project started before Jaxcalibur's write-up. Our gen-1 agents were on the ladder from August 2026, and the last one built before the write-up reached #5 on the gen-1 board.

The write-up showed that self-play PPO without search in training could reach #1 in generation 9. We then dropped search from training, as Wang (2024) had, and adopted several of its published ideas.

Our gen-9 agent's best on the same ladder was #5 (Section 4.5).

6.3 Other Pokémon agents

Earlier bots reported peak ranks of 8 (Wang, 2024) and 3 (pmariglia, 2025) on the gen-4 random-battle board. The PokéAgent Challenge, a NeurIPS 2025 competition (Karten et al., 2026), was played in generation 1 OU and generation 9 OU, where players build their own teams, on a separate server for AI agents, and reports no public-ladder rank.

7 Conclusion

MissingNo.Zero reached #1 on Showdown's random-battle ladder in generations 1, 2, 4, 6 and 7, using PPO self-play with no human data. We used the same algorithm in every generation and retuned it for each one.

References

Jaxcalibur (2026). *How Jaxcalibur works* (self-published write-up), https://jaxcalibur.github.io/, and "Jaxcalibur: A Gen 9 Randbats bot that reached #1 on the ladder", Smogon forums, https://www.smogon.com/forums/threads/jaxcalibur-a-gen-9-randbats-bot-that-reached-1-on-the-ladder.3787537/

Karten, S., Grigsby, J., Upaa Jr, T., et al. (2026). The PokeAgent Challenge: competitive and long-context learning at scale. NeurIPS 2025 Competition Track. arXiv:2603.15563, https://arxiv.org/abs/2603.15563; https://pokeagent.github.io

pmariglia (2025). *Foul Play: A Competitive Pokémon Showdown Battle Bot*, https://pmariglia.github.io/posts/foul-play/; https://github.com/pmariglia/foul-play

Schulman, J., Wolski, F., Dhariwal, P., Radford, A., Klimov, O. (2017). Proximal policy optimization algorithms. arXiv:1707.06347, https://arxiv.org/abs/1707.06347

Silver, D., Hubert, T., Schrittwieser, J., et al. (2018). A general reinforcement learning algorithm that masters chess, shogi, and Go through self-play. *Science* 362(6419), 1140–1144. https://doi.org/10.1126/science.aar6404

Wang, J. (2024). *Winning at Pokémon Random Battles Using Reinforcement Learning.* MIT MEng thesis, https://dspace.mit.edu/handle/1721.1/153888

Pokémon Showdown, https://github.com/smogon/pokemon-showdown; pkmn/engine, https://github.com/pkmn/engine; poke-engine, https://github.com/pmariglia/poke-engine; poke-env (Sahovic), https://github.com/hsahovic/poke-env

A The evidence

Each board at #1 (gen 9's at #5, its best), as the Wayback Machine captured it, and the evidence behind each run:

- Gen 1 (`nickgen1rbrlbot`): the board at #1, 2026-09-29, https://web.archive.org/web/20260929211638/https://pokemonshowdown.com/ladder/gen1randombattle?snap=1790716595. Evidence: Wayback captures, three timestamp proofs, screenshots.
- Gen 2 (`MissingNo.ZeroBot`): the board at #1, 2026-10-18, https://web.archive.org/web/20261018142133/https://pokemonshowdown.com/ladder/gen2randombattle?snap=1792333234. Evidence: Wayback captures, our hourly board snapshots, a timestamp proof, screenshots.
- Gen 4 (`nickgen4rbrlbot`): the board at #1, 2026-10-08, https://web.archive.org/web/20261008184118/https://pokemonshowdown.com/ladder/gen4randombattle. Evidence: Wayback captures, our hourly board snapshots, three timestamp proofs, screenshots.
- Gen 6 (`nickgen6rbrlbot`): the board at #1, 2026-10-16, https://web.archive.org/web/20261016121841/https://pokemonshowdown.com/ladder/gen6randombattle?snap=1792153118. Evidence: Wayback captures, our hourly board snapshots, a timestamp proof, screenshots.
- Gen 7 (`MissingNo.ZeroBot`): the board at #1, 2026-10-24, https://web.archive.org/web/20261024193431/https://pokemonshowdown.com/ladder/gen7randombattle?snap=1792870430. Evidence: Wayback captures, our hourly board snapshots, a timestamp proof.
- Gen 9 (`MissingNo.ZeroBot`): the board at #5, its best, 2026-10-25, https://web.archive.org/web/20261025072637/https://pokemonshowdown.com/ladder/gen9randombattle?snap=1792913158. Evidence: Wayback captures, our hourly board snapshots.

The other captures of each run (the account profiles, the peaks, the run ends), the git tags and file hashes, and their timestamp proofs are available from the author.